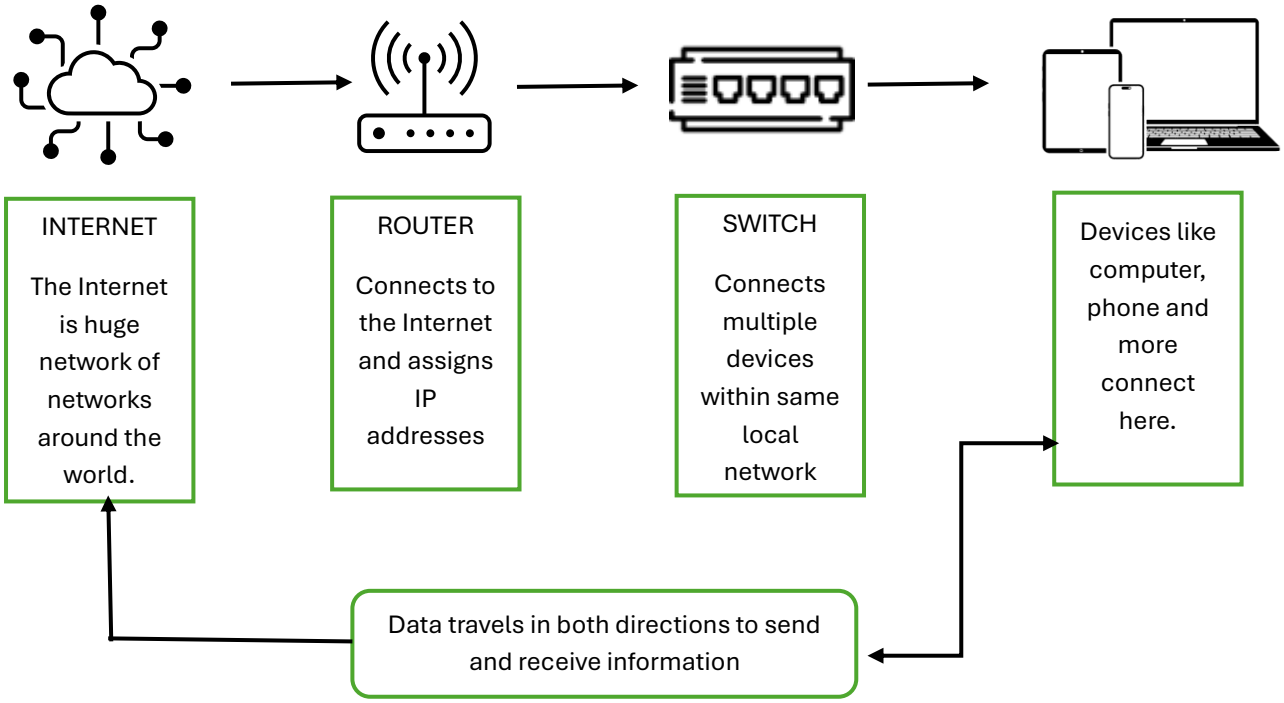
1. Draw and label a simple home network diagram showing a router, switch, and three devices (like a laptop, smartphone, and smart TV), indicating how each device connects to the network.



2. Create a comparison table listing at least four network devices (router, switch, hub, access point) and describe one real-world use case for each, referencing apps or services you use daily (e.g., WiFi for streaming on YouTube, LAN gaming, etc.).

Network Device	What It Does	Everyday Example
Router	Acts as the gateway to the outside world. It connects your house to the global internet and directs traffic.	Streaming Video: When you open YouTube or Netflix, the router fetches the video from the internet and sends it directly to your screen.
Switch	Acts as an intelligent power strip for internet cables. It lets you plug in multiple devices using cables and sends data *only* to the device that needs it.	Lag-Free Gaming: Plugging your PC or PlayStation into a switch for a super-fast, smooth multiplayer match on Steam without any wireless lag.
Hub	Acts as a blind broadcaster. It is an old, outdated version of a switch that copies data and sends it to *every* connected device, causing slowdowns.	Old Technology: Mostly replaced by modern switches now, but network technicians sometimes use them to test and monitor data traffic using apps like Wireshark.
Access Point	Acts as a Wi-Fi broadcaster. It takes a wired internet connection and turns it into a wireless signal.	Scrolling Social Media: Broadcasting Wi-Fi to a distant bedroom or backyard so you can scroll Instagram or TikTok smoothly on your phone.

3. Write a short scenario where you explain how data travels from your phone to the internet when you order food on Zomato, mentioning the role of each network device involved (router, switch, access point).

Imagine you are sitting on your couch using your smartphone to order dinner on the Zomato app. The moment you tap "Place Order," your digital request starts a high-speed journey through your home network to reach the internet.

- **Step 1: The Access Point (The Wireless Bridge)**
  Your smartphone isn't plugged into a wall, so it transmits your Zomato order through the air via radio waves. A Wireless Access Point inside your home catches this wireless signal and instantly converts it into physical electrical signals to travel down a network cable.
- **Step 2: The Network Switch (The Local Traffic Cop)**
  The cable from the access point plugs directly into a Network Switch. The switch acts like an intelligent mailroom. It reads the incoming digital package from your phone and immediately forwards it straight to your router, ensuring it doesn't collide with other household data (like someone else playing a game or watching a video)
- **Step 3: The Router (The Gateway to the World)**
  The data arrives at your Router. The router is the ultimate border checkpoint between your private home network and the massive public internet. It assigns a public return address to your request so Zomato knows where to send the confirmation. It then pushes your order out of your house, across the global internet infrastructure, and directly onto Zomato's servers.

Within a fraction of a second, the Zomato servers process your payment, and the confirmation data takes the exact same path in reverse—through your router, down the switch, out of the access point, and onto your screen with the message: *"Order Accepted!"*

4. Given this situation: you have a WiFi router, but your gaming PC only has an Ethernet port and is far from the router. Suggest two different network devices you could use to connect your PC to the internet and explain how each would work in this setup.

Hint: Think about devices like switches, powerline adapters, or WiFi extenders.

To connect your gaming PC to the internet without running a long cable across the house, you can use either a Powerline Adapter or a Wi-Fi Extender.

Option 1: Powerline Adapter Kit

A powerline adapter kit turns the existing electrical wiring inside your walls into internet cables.

- **How it works in your setup:** You get a matching pair of small plugs. You plug the first adapter into a wall outlet near your Wi-Fi router and connect them with a short Ethernet cable. Then, you plug the second adapter into a wall outlet near your gaming PC. Finally, you run a short Ethernet cable from that second adapter directly into your PC's Ethernet port.
- **The Result:** The internet data travels out of your router, through your home's electrical wires, and comes out right next to your computer, giving you a highly stable, physical connection.

Option 2: Wi-Fi Extender (with an Ethernet port)

A Wi-Fi extender catches the wireless signal from your router, boosts it, and shares it via a built-in physical plug.

- **How it works in your setup:** You plug the Wi-Fi extender into a wall outlet somewhere between your router and your gaming PC (where the Wi-Fi signal is still strong). The extender connects to your router completely wirelessly. You then plug a short Ethernet cable into the Ethernet port on the bottom of the extender and run it straight to your gaming PC.
- **The Result:** The extender acts as a wireless bridge. It catches the Wi-Fi from the air and feeds it directly into your PC's Ethernet port, making your computer think it is plugged directly into the network.